Predicting River Baseflow in the Republican River Basin with Multiple Linear Regression

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CS5830: Data Science in Practice

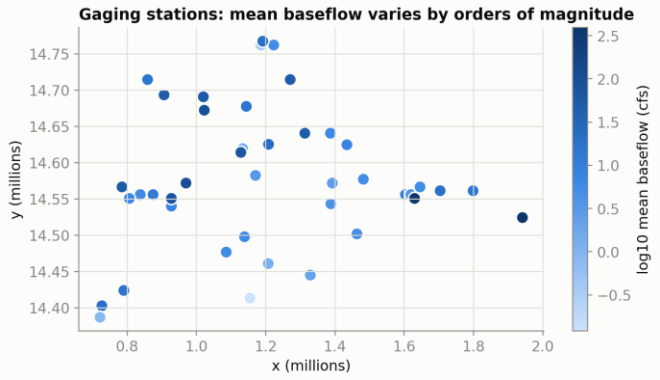
Introduction

Baseflow is the steady portion of a river that is fed by groundwater rather than by recent rain. It is what keeps streams flowing between storms and through dry summers, so it is the water that farmers, towns and wildlife depend on when conditions are hard. In the Republican River Basin, which spans parts of Colorado, Nebraska and Kansas, baseflow is also a legal matter: the three states share the river under an interstate compact, and disagreements over how groundwater pumping affects the river have ended up in court. Our analysis is meant to inform the people who manage and depend on this water: the state water agencies and compact administrators who divide it, and the farmers and communities who rely on it. How much of a river's baseflow can be explained by rainfall, plant water use and pumping? How well can the next decade's flow be predicted from earlier years? And what can the data, and what can it not, say about the effect of pumping?

We used the RRCA baseflow dataset, 15,591 monthly records for 46 river gaging stations from 1939 to 2000, and fitted multiple linear regression, a method that predicts a number from a weighted sum of inputs and shows how much each input matters. Rivers differ so much in size that knowing a river's own average already accounts for most of the variation. A model trained on 1939-1989 predicted the flows of 1990-2000 with an R^2 of 0.78, against 0.61 for simply guessing each river's average. Evapotranspiration (water returned to the air) lowers baseflow and precipitation raises it. At the ten stations with complete records, baseflow fell 38% while pumping grew, but the data cannot separate pumping from the general decline Please find the presentation slides and project repository linked here.

Dataset

The RRCA baseflow dataset comes from the groundwater model used by the Republican River Compact Administration to track how pumping affects the river. Each row is one river segment in one month, from September 1939 to August 2000: the month, the segment and the location of its gaging station (the instrument that measures flow), the evapotranspiration, precipitation and irrigation pumping in the area next to the segment, and the observed baseflow. The baseflow is not measured directly; it was estimated by splitting each river's total flow into surface-water and groundwater parts, and the file does not state its unit (we assume cubic feet per second, cfs). The data suits this question because it covers 61 years, includes the inputs that physical reasoning says should matter, and spans stations across the basin. Its main challenge is scale: a station's average baseflow ranges from 0.1 cfs at the smallest to about 390 cfs at the largest.



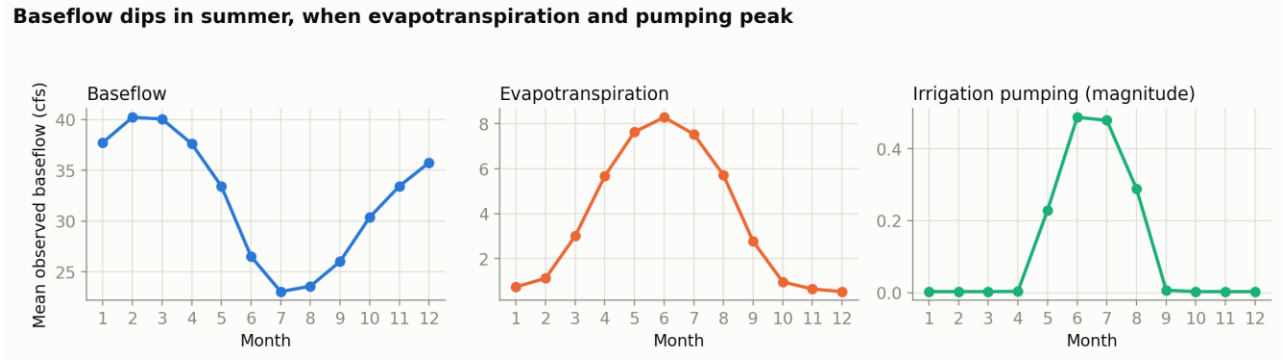
Gaging stations by location; darker means larger average baseflow (log scale).

Analysis technique

We used multiple linear regression, which predicts baseflow as a starting level plus a weighted sum of the inputs, one weight per input. We chose it because the question is not only "what will the flow be?" but "which factors matter, in which direction, and by how much?" Each weight answers that in plain terms (for example, how much a river's flow changes per unit of evapotranspiration), so a manager can check it against what they know of the river. It also matches the physical story: more rain, and less evaporation or pumping, should mean more groundwater reaching the river. We scored models with R^2, which measures how much of the variation in baseflow a model explains (1 is a perfect fit, and 0 means no better than always guessing the average), and with RMSE, the typical size of a prediction error in cfs. Its limit is that it assumes effects add up in a straight line, so we tested it carefully: we compared it with a baseline that ignores the weather, and we judged it on a later decade the model had not seen, not only on randomly chosen months.

Results

We see in the figure below that baseflow has a strong yearly cycle. Averaged over stations it is lowest in July (23 cfs) and highest in February (40 cfs), the reverse of evapotranspiration, and 99% of pumping falls from May to September. The biggest finding is how much rivers differ. A single line using weather, pumping and the date explains only about 12% of the variation ($R^2 = 0.12$), and no single input stands out when all rivers are plotted together (the strongest correlation with baseflow is only 0.27, for precipitation), because a station averaging 390 cfs and one averaging 0.1 cfs cannot share one line. Giving each river its own average level raises this to 0.78, but weather adds almost nothing to that on random months (0.785), so most of that apparent skill is just knowing which river is which.



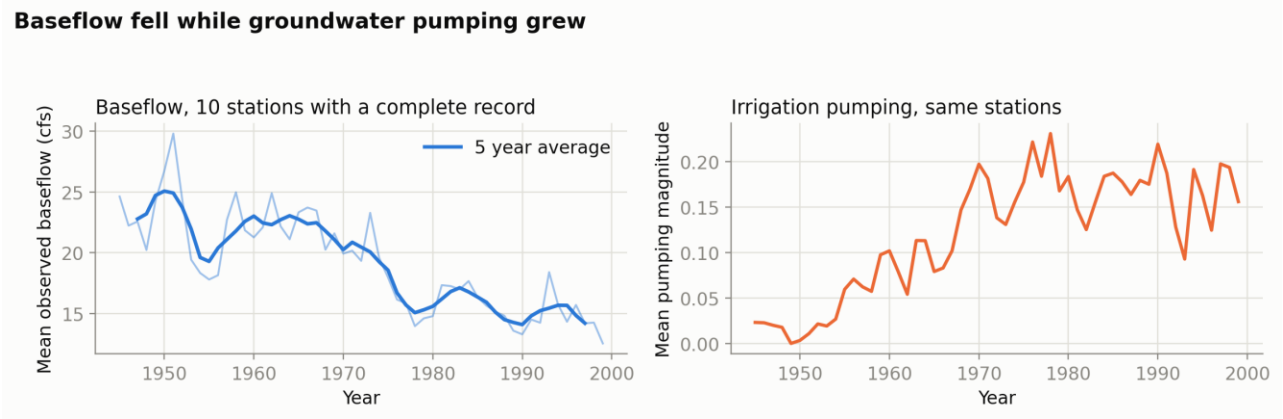
Average baseflow dips in summer, when evapotranspiration and pumping peak.

The fairer test is to learn from 1939-1989 and predict 1990-2000. Guessing each river's average scores 0.61. A model that predicts baseflow as a share of each river's normal flow, using the date, evapotranspiration, precipitation and pumping, scores 0.78 and cuts the typical error from 10.7 to 8.1 cfs. About three quarters of that gain comes from a steady downward drift over time. Evapotranspiration lowers baseflow and precipitation raises it, and the pattern holds river by river (precipitation in 81% of rivers, evapotranspiration in 69%). Across the ten stations with complete records, baseflow fell from about 23 cfs in 1945–49 to 14 cfs in 1995–99, a 38% drop, while pumping grew from almost none (Figure 3). Fitting a river on its own can work much better: the regression for segment 138 tracks 23 years of flow with an R^2 of 0.74 (Figure 5), though per-river fits vary widely (median R^2 0.45).

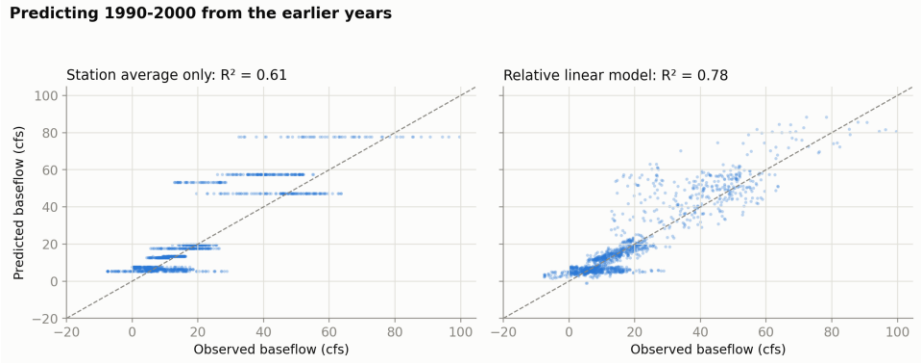
Takeaway: for a river with a record, a manager can forecast the next decade's flow reasonably well and flag stations running below what the weather predicts. But pumping and time rose together, so the data shows they moved together rather than proving that pumping caused the decline (once neighboring months are treated as related, the pumping effect cannot be told apart from zero), and a river with no history cannot be predicted from weather at all. The clearest limitation is the yearly cycle: a straight-line

model cannot reproduce seasonal swings that differ from river to river, and the month-of-year terms we tried did not help on 1990-2000.

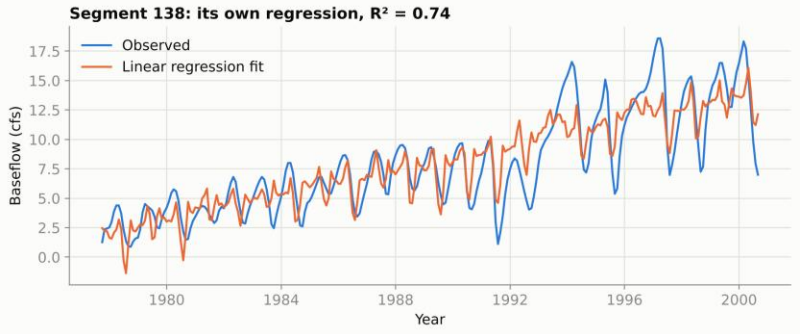
Model	R^2 random months	R^2 1990-2000	Typical error 1990-2000 (cfs)
Each river's own average only (no weather)	0.777	0.611	10.7
One line for all rivers: weather, pumping, date	0.095	−1.050	24.6
The same, plus each river's own level	0.785	0.697	9.5
Share-of-normal-flow model (our best)	**0.725**	**0.778**	**8.1**



At the ten stations with complete records, baseflow fell while pumping grew.



Predicting 1990-2000 from earlier years: each river's average only (left) versus the share-of-normal-flow model (right).



One river on its own (segment 138): observed baseflow and the fit of its own regression.

Technical

For the data preparation and analysis, date (days since 01/01/0000) minus 693963 gives days since 1900-01-01, converted to calendar dates. Segment_id is not a unique station: four ids (53, 152, 159, 171) have two x/y locations, which produces 393 duplicate segment-month rows, so we defined a station id from (Segment_id, x, y), giving 46 stations. Precipitation is never below 9.86 except in 4,416 rows that are exactly 0 (4,337 of them also have zero evapotranspiration, including seven stations with zeros in every month); we treated these as missing and left them out of every model that uses climate inputs, which leaves 11,175 rows from 37 stations. We kept the 232 negative baseflow values (222 from segment 178), which may reflect estimation error or a river losing water to the aquifer; dropping the zero-precipitation rows, the negative rows or both moves the best model's 1990–2000 R^2 only between 0.797 and 0.807. Precipitation repeats within a station for months (about 1.9 distinct values per station-year), so it behaves like an annual rather than a monthly input. Within each station and only across consecutive months, we built the mean precipitation of the previous 1, 3 and 12 months and the mean pumping of the previous 12 and 60 months, dropping rows without enough history (7,909 rows: 6,490 for training and 1,418 for testing, restricted to stations seen in training). We used ordinary least squares (scikit-learn LinearRegression; statsmodels for inference). Baseflow spans three orders of magnitude, so a single additive model is dominated by a few large stations; the "relative" model regresses Observed divided by the station's training mean (stations with a mean under 0.5 cfs are excluded from the fit) and multiplies the prediction back, which makes each effect proportional to a river's size. Residuals are strongly autocorrelated (median Durbin-Watson 0.33 in per-station fits), so ordinary p values are far too small; we report cluster-robust p values, clustered by station.

For the analysis process, our first models used a random 75/25 split of all rows: A (date, evapotranspiration, precipitation, pumping) $R^2 = 0.121$, B (A plus x and y) 0.131, and C (A plus Segment_id dummies) 0.815. The jump from A to C looked impressive, but a baseline that predicts only each station's average scored 0.78 on a comparable split, and a random split leaks neighboring months from the same river into both sets. We therefore followed Xu and Valocchi (2015) and trained on 1939-1989 and tested on 1990-2000, where the baseline falls to 0.61, the additive station-level model reaches 0.70 and the relative model 0.78 (date alone 0.74; weather and pumping without the date 0.67). Failed or marginal attempts: coordinates add nothing beyond the station level; month-of-year dummies improved the random split (0.785 to 0.793) but hurt the future test (0.697 to 0.675; relative model 0.778 to 0.750), suggesting the seasonal pattern changed as pumping grew; lagged precipitation and pumping gave the additive model a small gain (0.697 to 0.721) and the relative model none (0.776); a weather-only model fails on stations it never saw (leave-stations-out R^2 per fold from 0.09 down to -54, mean -11), because a new river's size is not in the inputs; and separate per-station regressions (at least 30 months each, with pumping dropped at the three stations where it is zero throughout) had a median R^2 of 0.45. The pumping coefficient has the expected sign (clustered $p \approx 0.28$) but cannot be isolated from the time trend with ten complete-record stations. Alternatives we did not try: nonlinear models such as the random forests and support vector regression of Xu and Valocchi, mixed-effects models with random slopes per station, harmonic (sine and cosine) seasonal terms or seasonal coefficients that change by decade, and a log transform of baseflow (complicated by the negative values).

Reference

Xu, T. and Valocchi, A. J. (2015). Data-driven methods to improve baseflow prediction of a regional groundwater model. Computers & Geosciences 85, 124–136.